

UNIVERSITY OF BARISHAL

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

SOFTWARE ENGINEERING PROJECT PROPOSAL

GroupSync – Intelligent Group Project Management with Contribution Analytics

CSE-3104: Software Engineering and Information System Design Lab

Submitted By:

Team SlackerSlayer

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Submitted To:

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Submission Date: 20th May 2026

Formation of Team SlackerSlayer

Our project team consists of six members who will work together throughout the development process of GroupSync. The team has been organized in a way that ensures proper task distribution, collaboration, and efficient project management. Each member has specific responsibilities based on their skills and interests, although all members will contribute to both frontend and backend development during different stages of the project.

Serial	Name	Role
01	Humayra Alamgir Nuha	Team Leader, Backend development
02	Puja Gain	Requirement analysis , database design,backend development
03	Moumita Ghose	Front-end developement and UI/UX design section
04	Ritu Akter Samia	Front-end development and documentation
05	Sanjida Akter Mim	SQA Engineer , backend development, backend support
06	Maimuna Akter	Frontend development,deployment, maintenance tasks.

SWOT Analysis

Strengths The team has a good understanding of full-stack web development technologies, especially the Php + Laravel. Team members are familiar with GitHub collaboration and responsive web design. The project also benefits from proper role distribution and teamwork through Agile development practices.	Weaknesses As undergraduate students, our team has limited experience working with large-scale, real-time systems. Managing academic workload alongside project deadlines may also create challenges during development. In addition, the project depends on third-party APIs such as GitHub integration services.
Opportunities The project provides an opportunity to solve a common academic problem faced by universities during group project evaluation. It will also help team members gain practical experience in software engineering, teamwork, and full-stack development.	Threats Possible API limitations, hosting downtime, and security vulnerabilities may affect system performance. There is also a risk of biased peer reviews, which may influence contribution evaluation accuracy.

Idea Generation

Identify Domain:

Education (University Group Project Management)

Observe Real World Problems:

In university group projects, contribution imbalance is a common issue — some students work hard while others remain passive, yet all receive equal grades. GitHub alone cannot track attendance or teamwork, and manual peer reviews are often biased.

Why GroupSync was selected:

- **Feasibility:** Feasible with proper planning. We have 6 team members and 4 weeks. PHP stack is well-documented. GitHub API is free.
- **Impact:** Solves the real problem of unfair grading in group projects. Teachers can finally see individual contribution levels. Hardworking students feel recognized. Free-riders are identified early. Promotes accountability and fairness across the entire university.
- **Innovation:** While each individual feature exists separately, combining GitHub activity, attendance, peer reviews, working hours, and task completion into a single weighted score is unique. No existing academic tool does all of this in one platform.
- **Scope:** Any university course that uses group projects can adopt GroupSync. It works for coding projects (via GitHub), research projects, business plans, design projects, and more. The system is scalable to hundreds of courses and thousands of students.
- **Learning Value:** Team members will gain hands-on experience in: full-stack web development, REST API integration (GitHub API), database design and schema modeling, weighted algorithm development, user authentication and security, responsive UI/UX design (TailwindCSS), Agile methodology and sprint planning, team collaboration using Git and GitHub, deployment on cloud platforms (Vercel, Railway, infinityFree), and project documentation and presentation.

Overview

GroupSync is a web-based platform designed to improve collaboration and fairness in university group projects. In many academic projects, it becomes difficult for teachers to identify the actual contribution of each student. As a result, some students complete most of the work while others receive equal marks despite limited participation.

To address this issue, GroupSync provides a centralized system where students can manage group activities, track project progress, and monitor contributions. The platform integrates GitHub activity, attendance records, working hours, and peer reviews to generate a contribution score for each group member.

The system includes separate dashboards for students, teachers, and administrators. Teachers can monitor group performance and generate evaluation reports, while students can track their own activities and participation records.

The proposed system is mainly intended for academic use and focuses on improving transparency, accountability, and fairness in collaborative university projects.

Problem Statement

Group projects are commonly used in universities to develop teamwork and collaboration skills among students. However, evaluating individual performance in group assignments is often difficult for teachers.

In many cases, a small number of students complete most of the project work, while some members contribute very little. Despite the unequal participation, all students usually receive the same marks. This creates dissatisfaction among active students and reduces accountability within the team.

Although tools like GitHub can track code commits, they cannot fully measure other important activities such as communication, attendance, teamwork, and task participation. Manual peer evaluations are also not always reliable because they may be biased or incomplete.

Therefore, there is a need for a system that can combine different contribution factors into one platform and help teachers evaluate students more fairly and efficiently.

Motivation

The motivation behind GroupSync comes from real experiences during university group projects where workload distribution was often unequal. This created unfair evaluation situations and reduced accountability among students.

This problem inspired us to develop a system that promotes fairness and transparency in academic collaboration. GroupSync allows teachers to monitor contributions more effectively while encouraging students to remain active in group work.

The project also provides us with practical experience in full-stack development, database design, authentication systems, API integration, and teamwork in software engineering.

Additionally, it helps us understand how real-world software systems are designed and implemented to solve practical problems in educational environments.

Expected Features and Requirements

The following tables outline the core features of the GroupSync platform.

Student Features:

Features	Description
User Registration & Login	Students can register using their university email address and securely log in to access their personalized dashboard.
Profile Management	Students can update their personal information, upload a profile picture, and change their password.
Create Group	Any student can create a new group and receive a unique invitation code to share with team members.
Join Group	Students can join an existing group by entering the invitation code provided by the group creator.
Connect GitHub Repository	Students can connect their GitHub account and link the group's repository for automatic activity tracking.
View Dashboard	Students can see their personal contribution score, group progress, and individual performance metrics on a visual dashboard.
Submit Peer Review	Students can submit anonymous peer reviews for each team member based on communication, reliability, and task participation.
View Own Contribution History	Students can view a breakdown of their GitHub commits, attendance records, logged hours, and peer review scores.
Receive Notifications	Students receive alerts when peer reviews are pending, when their contribution score drops, or when a teacher leaves feedback.

Professor / Teacher Features:

Feature	Description
Teacher Registration & Login	Register by using their institutional email and log in with secure authentication
Create Course	creating new courses with details such as course code, title, semester, and enrollment period.
Enroll Students	adding students to a course manually or allow self-enrollment using a course code

Create Group Assignments	Creating group project assignments and set evaluation parameters (weightage of each metric).
View All Groups	Seeing a list of all groups within a course along with group size and completion status.
Monitor Individual Contributions	Viewing each student's contribution score (0-100) and a detailed breakdown of how the score was calculated.
Generate Reports	Generating downloadable reports (PDF/Excel) of contribution data for final grading.
View Analytics Dashboard	Seeing visual charts and graphs showing contribution distribution, low performers, and group comparisons.
Identify Low Contributors	The system automatically flags students whose contribution score falls below a teacher-defined threshold.
Leave Feedback	Teachers can send feedback or warnings to individual students or entire groups.

Admin Features:

Features	Description
Admin Login	Secure admin panel login with role-based access control.
Manage Users	Admin can view, add, edit, or delete student and teacher accounts.
Monitor Platform Activity	Admin can view system logs, active users, and overall platform usage statistics
System Settings	Admin can configure system-wide settings such as email verification requirements, session timeouts, and backup schedules
Handle Support Requests	Admin can view and respond to user support tickets.

Development Approach

The project will follow the Agile Software Development Model. Development will be completed through multiple small iterations or sprints. Each sprint will focus on a specific module of the system, allowing continuous testing and improvements throughout the development process.

The Agile approach has been selected because it supports flexibility, teamwork, and faster development. Regular feedback and testing will help identify problems early and improve system quality.

Development Model

The Agile model is selected for this project because it allows iterative development and continuous improvement. It supports collaboration among team members and helps in adapting to requirement changes during development.

Each sprint will deliver a functional module of the system, ensuring steady progress and early detection of issues.

Tools and Technology

Component	Technology
Frontend	HTML5 , CSS (Taliwind CSS), Javascript ,React.js
Design & Prototyping	Figma for wireframes, UI prototypes, and design system documentation
Backend	PHP (Laravel),Node.js with Express.js — REST API architecture
Database	MySQL-for structured data storage
GitHub Integration	GitHub REST API for fetching commit history, repository activity, and push frequency
Version Control	Git with GitHub — all 6 team members added as collaborators with branch protection rules
Deployment	InfinityFree
Project Management	Trello for sprint planning, task assignment, and progress tracking
API Documentation	Postman for API endpoint documentation and team collaboration

Project Timeline

Week	Planned Activities
Week 1	Requirement analysis and system planning, Database setup and authentication system
Week 2	Group management module development , GitHub integration and repository connection
Week 3	Contribution tracking and scoring system , Peer review system and dashboards

Week 4	Analytics and report generation , Testing, deployment, and documentation

Software Development Life Cycle (SDLC)

The project follows a structured SDLC process to ensure proper development and maintenance.

Requirement Analysis: User needs and system requirements are collected and analyzed.

System Design: Architecture, database schema, and UI design are prepared.

Implementation: Frontend and backend development are carried out using selected technologies.

Testing Unit: testing, integration testing, and system testing are performed to ensure quality.

Deployment: The system is deployed on cloud platforms for real-world access.

Maintenance: Continuous updates, bug fixes, and improvements are applied after deployment.

Budget and Estimation

Category	Description	Estimated Cost (BDT)
Development and Implementation	Backend, frontend, integration, and system development activities	450,000
UI/UX Design and Prototyping	User interface design, wireframing, and usability improvements	120,000
Cloud Hosting and Deployment	Server hosting, domain, SSL, and deployment services	140,000
Database and Storage Services	MongoDB Atlas, cloud storage, backup, and maintenance	100,000

API and Analytics Services	GitHub API integration, analytics processing, and testing	160,000
Security and System Testing	Authentication testing, bug fixing, and performance evaluation	130,000
Software Tools and Collaboration	Development tools, project management, and version control services	90,000
Documentation and Presentation	Printing, report preparation, and presentation materials	40,000
Internet and Technical Support	Internet expenses, troubleshooting, and support services	70,000
Research and Consultation	System planning, architecture review, and technical consultation	110,000
Maintenance and Future Improvements	System updates, optimization, and post-development maintenance	90,000
Contingency Cost	Unexpected development and operational expenses	100,000
Total Estimated Budget		1,500,000 BDT

Conclusion

GroupSync is a web-based academic project management system that aims to improve fairness and accountability in university group work. By combining contribution tracking, peer reviews, attendance monitoring, and GitHub activity, the system helps teachers evaluate student participation more effectively.

The project also provides students with a more organized way to manage collaborative tasks and monitor their own progress. Through this project, the team will gain practical experience in software engineering, full-stack development, database design, and collaborative system development.

In the future, additional features such as real-time notifications and advanced analytics can be added to further improve the platform and make it more effective for academic project management.